Shivam Patel

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Summary

PhD researcher in scientific ML, numerical methods, and optimization, with experience in large-scale data analysis on HPC systems. Proficient in Python, C++, and Linux-based scientific computing environments with hands-on experience developing and optimizing computational algorithms, validating numerical accuracy against analytical solutions, and building efficient ML pipelines in PyTorch.

Education

PhD | Computational and Applied Mathematics		
Portland State University	GPA: 4.00/4.00	Sep 2024 - Present
MS | Mathematics & Data Science		
Indian Institute of Science Education and Research, Bhopal	GPA: 9.01/10.00	Aug 2023 - May 2024
BS | Mathematics & Data Science		
Indian Institute of Science Education and Research, Bhopal	GPA: 8.78/10.00	Aug 2019 - May 2023

Skills

Programming Languages & Tools: Python (Seaborn, Matplotlib, Plotly, PyTorch, TensorFlow, Scikit-Learn, NumPy, SciPy, Pandas), PowerBI, Git, GitHub, R, SQL, C++, MATLAB, C, Tableau, High Performance Computing on Linux Clusters

ML/AI Techniques: Deep Learning, Image Classification, Time-Series Analysis, Reinforcement learning, Physics-Informed Neural Networks (PINNs), Operator Learning, Neural Operators, Transformers, LLMs, Statistical Data Analysis

Scientific Computing Software: MFEM, Netgen/NGSolve, Mathematica, DeepXDE, CUDA

Others: Numerical analysis, Scientific computing, Numerical Linear Algebra, Finite Element Methods (FEM), Model-constrained Optimization, Optimization Algorithms, Reinforcement Learning, Econometrics, Mathematical Modeling, Image processing

Experiences

Graduate Research Assistant | Portland State University — Sep 2024 - Present

- Developing new method for solving **eigenvalue problems of magnetic Schrödinger operators**, which helps to understand the energy quantization of quantum particles under the influence of an electromagnetic field.
- Generated new **quadrature methods for singular integral kernels**, utilizing **HPC Clusters** for large-scale computations.
- Utilized advanced finite-element frameworks such as **NGSolve** and **MFEM** to compute eigenvector localization, with applications to wave localization in disordered media.
- Built and trained **PINNs** and **Fourier Neural Operators** in PyTorch to solve PDE-governed electromagnetic wave propagation problems, reducing reliance on traditional numerical solvers.

Financial Data Analyst | Dark Horse Stocks, India — May 2024 - Aug 2024

- Utilized SQL and Python to automate data collection from BSE and NSE, reducing manual data entry time by 30%.
- Presented detailed financial analysis using statistical models to forecast portfolio performance, and evaluate the impact of investment on various financial scenarios to enhance clients' understanding and confidence in investment choices.
- Maintained dynamic Power BI dashboard to produce weekly market insights and investment recommendations, contributing to an 8% increase in client engagement.

Graduate Research Assistant | IISER Bhopal — Aug 2023 - Apr 2024

- Developed a full **FEM** pipeline in MATLAB and C++ to solve Laplace, Poisson, and Helmholtz equations, validated against analytical solutions with convergence rates consistent with theoretical $O(h^2)$ error bounds.
- Applied the Galerkin Method to approximate weak solutions and convergence analysis for robust multi-physics simulations.
- Analyzed existence and uniqueness of weak solutions in Sobolev spaces, forming the theoretical backbone for a robust solver.

Research Intern | University of Calgary, Canada — May 2023 - Jul 2023

- Worked on functional-analytic concepts of Banach and Hilbert spaces to understand the **Riesz Representation Theorem** and the **Lax-Milgram Theorem**, focusing on their applications in PDE theory.
- Analyzed the existence and uniqueness of weak solutions to elliptic PDEs by studying Sobolev Spaces and their properties.
- Worked on the Laplace, Heat, and Wave Equations, exploring their physical applications and the **continuous Galerkin FEM** for solving boundary value problems.

Teaching Experiences

Graduate Teaching Assistant | Portland State University — Sep 2025 - Present

- Independently delivered lectures and led recitations for a Calculus section of 30 students, taking full ownership of instruction and student engagement.
- Conducted regular office hours and one-on-one tutoring sessions, providing targeted support to strengthen student understanding of core concepts.
- Evaluated student performance through grading exams and assignments, providing feedback to drive improvement.

Conference Presentations and Posters

- **Shivam Patel**, Jeffrey Ovall, Li Zhu. Accurate Computation of Eigenvalues of the Magnetic Dirichlet Laplacian via Boundary Integral Methods. 5th Biennial Meeting of the Pacific Northwest Section of SIAM, Oct 2025
- **Shivam Patel**, Jeffrey Ovall. Eigenvalues of Laplacian on perforated domains. 9th Cascade RAIN Meeting, Apr 2025

Relevant Projects

Neural Operator-based Solver for Heat Transfer Problems | Portland State University

- Formulated a **Fourier Neural Operator (FNO)** in PyTorch to learn the solution operator of the heat equation, achieving an MSE of 10^{-5} - competitive with high-resolution traditional numerical solvers.
- Designed a custom training methodology enabling the model to generalize across varying initial conditions and complex non-linear physical systems.
- Demonstrated **zero-shot super-resolution**, generating high-resolution outputs beyond the training grid without retraining.
- Established a rigorous mathematical framework for **operator learning**, based on functional analysis & approximation theory.

Physics Informed Neural Networks for solving differential equations | Portland State University

- Designed and trained PINNs in PyTorch to solve boundary value problems involving Laplacian eigenfunctions .
- Implemented **PDE-based loss functions** and optimized training using the **L-BFGS algorithm**, significantly accelerating convergence over standard gradient descent.
- Applied **Runge-Kutta time** stepping within a discrete-time PINN framework to solve transport equations.
- Validated the PINN framework across multiple initial conditions with an accuracy ranging from 90% - 95%.
- Analyzed network architectures, activation functions, and loss balancing to improve training stability and generalization.

Predicting Heart Disease using classification models | IISER Bhopal

- Conducted **exploratory data analysis (EDA)** to uncover distributions, class imbalances, and feature correlations using **Chi-Square tests**, for cardiovascular risk modeling.
- Developed and optimized 6 classification models (**Logistic Regression, k-NN, Random Forest, SVM, Bagging, Boosting**) with cross-validation and hyperparameter tuning.
- Achieved best performance with **non-linear SVM**, attaining 80% recall and 71% F1-score.
- Identified **linear SVM** as the most efficient alternative (78% recall, 72% F1-score), providing a clinically interpretable model.

Task Allocation, Sequencing and Scheduling Problem | IISER Bhopal

- Formulated a combinatorial optimization problem to minimize total mission time for a human-robot collaborative team operating across multiple target locations.
- Implemented **Genetic Algorithms (GA)** in Python to efficiently explore large solution spaces, outperforming brute-force permutation approaches on complex task configurations.
- Validated performance through computational simulations, demonstrating consistent improvements over baseline approximation algorithms across diverse scenarios.

Weather Image Classification | Portland State University

- Preprocessed and augmented uncurated weather image dataset using resizing, normalization, and random transformations.
- Built a **custom CNN in PyTorch** with batch normalization, achieving a test accuracy of 67.37% on a challenging near-indistinguishable class distribution.
- Applied **transfer learning with ResNet-18**, boosting test accuracy to 80.19%, showing a significant improvement.
- Performed hyperparameter tuning on dropout rates and learning rates to generalize across ambiguous visual categories.

Noise Driven Route Planning using Genetic and A* Algorithms | IISER Bhopal

- Collected and processed real sound samples using **Short-Time Fourier Transform (STFT)** via **Librosa** for noise feature extraction, filtering, and normalization.
- Implemented **Genetic** and **A* algorithms** in Python to generate optimal routes balancing noise exposure and travel distance.
- Developed a noise characterization model integrating noise levels into loss function, enabling noise-aware path optimization.

Relevant Coursework

At PSU: Advanced Optimization and Data Assimilation, Statistical Learning, Advanced Numerical Analysis, Functional Analysis

At IISER Bhopal: Applied Optimization, Combinatorics & Graph Theory, Applied Deep Learning, Machine Learning, Advanced Programming in Python, Probability & Statistics, Econometrics, Data Structures & Algorithms, Numerical Analysis, Artificial Intelligence, Signal & Systems, Fourier Analysis, Discrete Mathematics, Partial Differential Equations

Coursera: Fibonacci Numbers & Golden Ratio, IBM Machine Learning Specialization, Deep Learning Specialization

Accomplishments

- Participated in **CBMS Conference: Research at the Interface of Applied Mathematics and Machine Learning** at the University of Houston in Dec 2025
- Participated in **Structure-Preserving Scientific Computing and Machine Learning Summer School and Hackathon** organized by PIMS Collaborative Research Group at the University of Washington in Jun 2025
- Presented at **Advanced Instructional School** organized by the National Centre for Mathematics at IISc Bangalore in Dec 2023
- Attended **Vijyoshi 2019 – National Science Camp** organized by Kishore Vaigyanik Protsahan Yojna (KVPY) at IISER Kolkata
- Recipient of **DST-INSPIRE Scholarship** issued by Department of Science and Technology, Government of India